

Toy Polynomial Kyber (3x3)

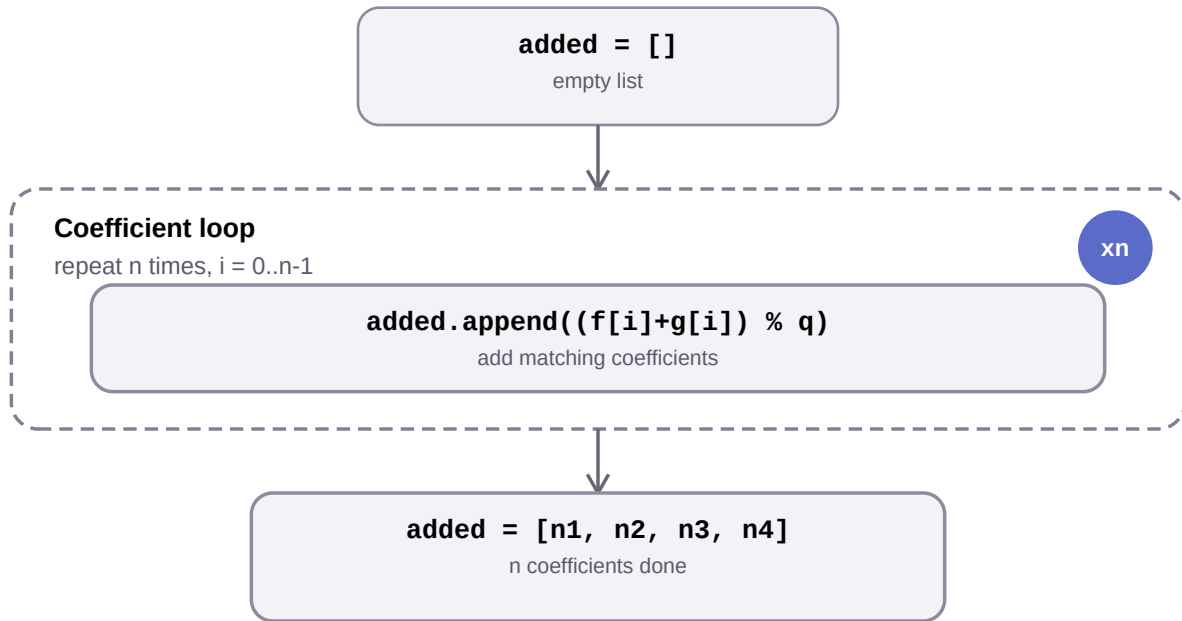
The polynomial ring: `poly_add`, `poly_sub`, `poly_mult`

This document accompanies `toy_poly_kyber_3x3.py`.

It covers the three new building-block functions that let every 'number' in the toy integer version become a polynomial instead:
`poly_add`, `poly_sub`, and `poly_mult`.

`KeyGen`, `Encapsulate` and `Decapsulate` themselves are not repeated here - they use the exact same structure as the integer version, just with these three functions standing in for `+`, `-` and `x`.

poly_add(f, g)

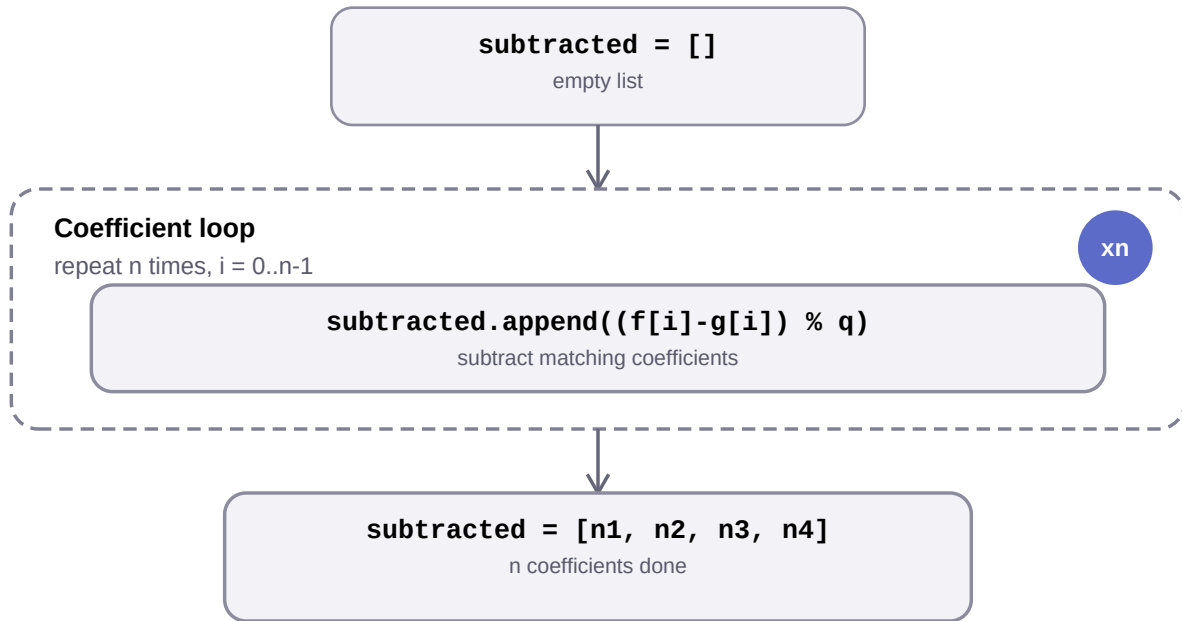


Same shape as vector addition from the integer version.

Match up `f[i]` with `g[i]` at every position, add them, wrap mod `q`.

Nothing about polynomials changes this - only what the numbers represent (coefficients, not standalone values).

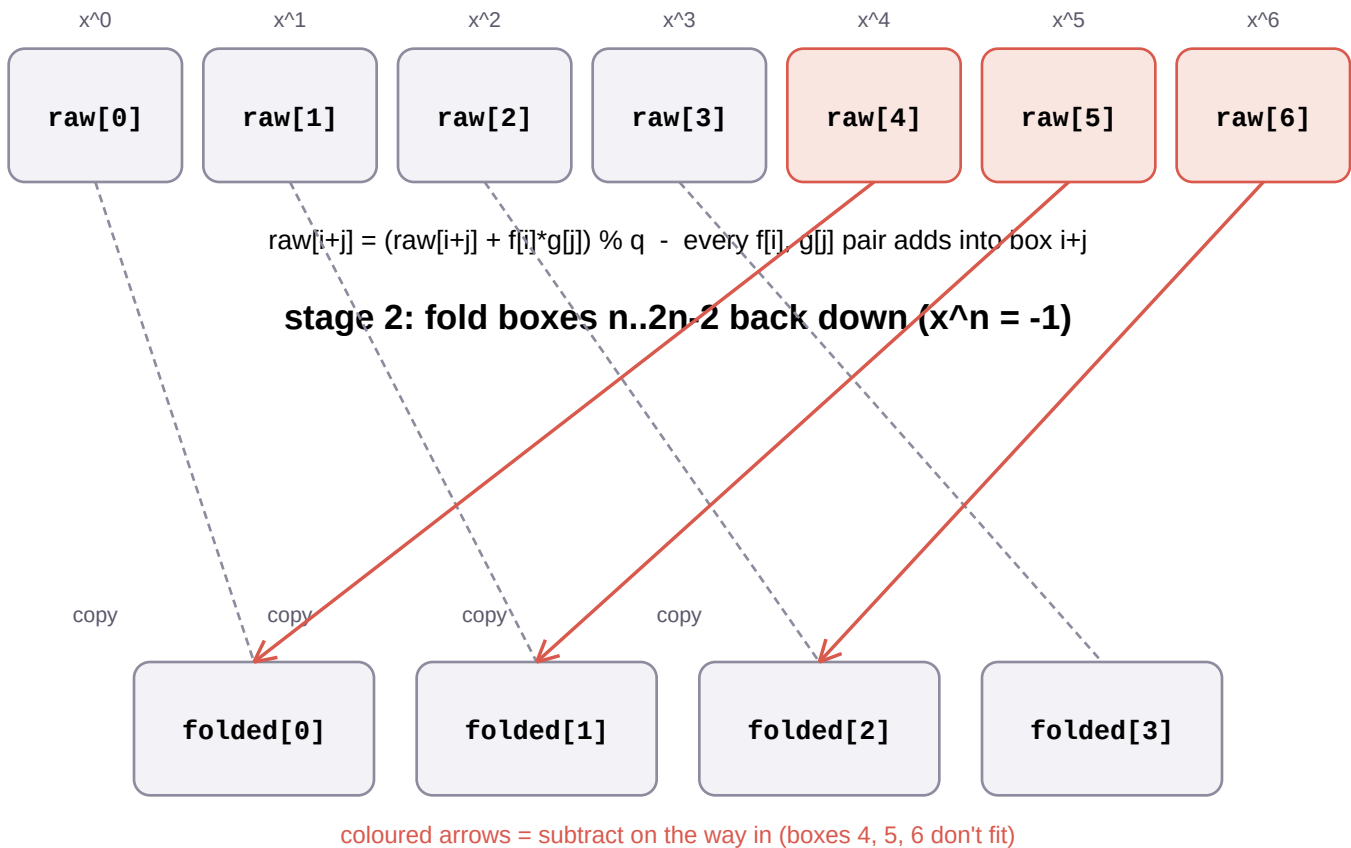
poly_sub(f, g)



Identical to `poly_add` - only the operator changes, `+` becomes `-`.

Needed for Decapsulate's $w = v - S_t \cdot u$, since Python cannot subtract two lists directly with a plain minus sign.

poly_mult(f, g) - stage 1: multiply into raw



Two stages: (1) multiply every pair of terms, drop each product into the box matching its power of x ($i+j$); (2) fold anything at position n or higher back into range $n-1$, subtracting instead of adding, because $x^n = -1$ flips the sign.

Verified against 2000 random inputs, checked against an independently-written reference calculation (`verify_poly_mult.py`).

KeyGen, Encapsulate, Decapsulate - what changed

Every step - building $A/S/e$, computing b , $r/e_one/e_two/m$, A_t , u , v , S_t , w , and decoding m - uses the exact same loop structure as the integer version (`toy_integer_kyber_3x3.py`). Only three things changed throughout:

1. Every plain number became a polynomial (a list of n coefficients).
2. Every $+$ between two of these became `poly_add`.
3. Every $*$ between two of these became `poly_mult` (and the one - in Decapsulate's w calculation became `poly_sub`).

The 'shape' of the algorithm - which loops, in what order, what gets added to what - never changed. Only the operations inside did, because lists can't use $+$, $-$, $*$ directly the way numbers can.